

Judah Sleibi

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SUMMARY

AI Engineer specializing in NLP and LLM-based systems, with hands-on experience building AI-driven applications using FastAPI, Hugging Face, and OpenAI APIs. Currently pursuing a Master's in Artificial Intelligence, with a strong foundation in Python, machine learning, and data processing. Published IEEE researcher focused on developing explainable, production-oriented AI systems bridging research and real-world applications.

TECHNICAL SKILLS

Machine Learning & AI:

- PyTorch, TensorFlow, Transformers (mBERT), Hugging Face, supervised learning, model fine-tuning, inference pipelines.

LLMs & Generative AI:

- OpenAI API (GPT), Gemini API, prompt engineering, conversational AI systems, context handling.

Data Analysis & Visualization:

- NumPy, Pandas, Matplotlib, Seaborn.

Backend & APIs:

- FastAPI, Pydantic, REST API development.

Databases:

- SQL, PostgreSQL.

Computer Vision:

- OpenCV, Pillow.

Tools:

- Git, GitHub, Docker.

TECHNICAL PROJECTS

SparkVision AI Agent - Real-Time Child Digital Eye-Strain Detection IEEE Publication | 2025 | [Link](#)

- Built real-time CNN + rule-based system for eye-strain detection (98.82% accuracy)

Early prediction of kidney dysfunction in diabetes patients IEEE Publication | 2025 | [Link](#)

- Built ML models (KNN, Random Forest) for early prediction using clinical features (~98% accuracy)
- Applied preprocessing and threshold-based labeling on 499 samples (13 features)
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Arabic Voice Assistant – AL NAYZAK | 2026 | [Link](#)

- Built a real-time Arabic voice assistant with WebSocket streaming, VAD, and STT/LLM/TTS pipeline
- Deployed full-stack on Hugging Face Spaces using Docker with 24/7 uptime monitoring
- Integrated Google Gemini 2.5 Flash with a PDF knowledge base for domain-specific Arabic Q&A

Phishing Detection Ensemble (Logistic Regression + Char-CNN) | 2026

- Built hybrid ML/DL model achieving 0.9857 ROC-AUC (test)
- Trained Char-CNN with early stopping, reaching ~0.9936 validation ROC-AUC

LLM-Powered Conversational AI System

- Built GPT-based chatbot using OpenAI API + FastAPI with multi-turn context handling
- Containerized with Docker and structured APIs using Pydantic

Multilingual Toxicity Dataset for Arabic–English–Arabizi Text

- Built large-scale Arabic–English–Arabizi dataset with span-level annotations for NLP tasks

Transliteration-Aware Toxicity Detection (NLP)

- Built multilingual NLP pipeline (mBERT + XGBoost) for Arabic–English–Arabizi toxicity detection (>0.99 F1)
- Designed transliteration-aware preprocessing and integrated SHAP/LIME for explainability

Ibtikar | [Link](#)

- Built mBERT-based mobile app for toxicity detection on social media (>99% accuracy)

CerebroX – Automated Data Pipeline & EDA Platform | [Link](#)

- Built automated pipeline for CSV/XLSX → EDA reports with preprocessing, visualization, and SQL integration.

Rule-Based Arabizi-to-Arabic NLP Engine

- Built rule-based system for Arabizi → Arabic conversion using linguistic patterns

FastAPI Product API | [Link](#)

- Built a RESTful API using FastAPI, Pydantic, SQLAlchemy, and PostgreSQL implementing CRUD operations; managed with Git/GitHub.

EDUCATION

Data Analyst| Bootcamp at Spark x Udacity – Palestine 1-2-2025 | 2-6 -2025

- Pursuing an intensive data science scholarship focused on Python (NumPy, Pandas), data visualization (Matplotlib), and machine learning, with hands-on projects, expert mentorship, and career coaching.

Quality Assurance and software testing | Bootcamp at AXSOS Academy | Palestine 1-10-2024 | 1-2-2025

- Completed a 16-week, 500-hour QA bootcamp, focusing on Manual Testing, Back-end Testing, Automation, Mobile Testing, and advanced automation techniques, hands-on project and team work dynamics.

Arabic American University | Master, Artificial Intelligence | Ramallah, Palestine. 1-10-2024 | Present

Birzeit University | Bachelor, Mechatronics Engineering| Birzeit, Palestine. 1-10-2018 | 1-2-2024

Introduction to Machine learning | 40 hours training| Birzeit university, Palestine. 1-5-2022 | 1-7-2022

PROFESSIONAL EXPERIENCE

Quality assurance engineer | Pal-Swiss Tech Hubs | Ramallah 1-3-2025 | 30-10-2025

- Automated data validation pipelines using Python and SQL to ensure data integrity across systems
- Built and maintained automated testing workflows (Cypress), reducing manual QA effort by 50%
- Analyzed backend data using SQL to validate system outputs and detect inconsistencies
- Contributed to improving system reliability through structured testing and data-driven validation

MEAL Coordinator | Al-Nayzak | Ramallah 1-11-2025 – Present

- Applied machine learning models (Logistic Regression, Neural Networks) to analyze program performance and predict risks
- Built Python-based data pipelines for data cleaning, preprocessing, and analysis
- Conducted clustering and statistical analysis to segment performance metrics and identify optimization opportunities
- Generated data-driven insights to support decision-making and operational improvements

AI & Data Science Instructor & Content Developer | Digital Academy of Al Nazyal 1-2-2026 – Present

- Delivered 300+ hours of training in Data Science, Machine Learning, and Generative AI
- Taught Python, SQL, and ML concepts including neural networks and LLMs
- Designed hands-on projects covering real-world ML and AI applications
- Developed curriculum and training materials for applied AI learning